

fcompress: transport gathering and canonical packing

local-mixing project notes

2026-09-05/06

Code: `src/postprocessing/compress.rs`, `src/postprocessing/downhill.rs`,
`src/postprocessing/bin/fcompress.rs`, `src/engine/format.rs` (anfl).
Manual: `docs/POSTMIX_MANUAL.md` §3.
Markdown twin: `docs/FCOMPRESS_TRANSPORT_AND_PACKING.md`.

1 Summary

fcompress is the attacker-computable final pass of the GSS pipeline and the honest effective-size evaluator of a mixed circuit. Three new gathering rules make it remove far more than before, and a packing pass now spells the result in a canonical form.

circuit	gates in	old pass (2026-08-22)	new pass	literals
A1 fmix benchmark (256 wires)	607,365	575,396 (94.7%)	420,279 (69.2%)	55.3%
K2 hold-10 delivered final (512 wires)	3,748,210	fixed point (99.9%)	2,165,945 (57.8%)	47.7%
balanced K2 2-eff final	3,067,544	fixed point	1,614,661 (52.6%)	43.2%
K2 hold-30 2-eff final	3,053,636	fixed point	1,607,266 (52.6%)	43.3%

Every number was checked by **fcompress**’s own sampled global check (64 rounds $\times$ 64 lanes on all wires) and by an independent evaluator sharing no code with it, on 2,048 fully random inputs over all wires. Consequences: effective sizes and “residual $\gtrsim 90\%$ ” figures quoted before 2026-09-05 are about $1.7\times$ too high; the crossing stage’s $\times 1.8$ expansion is mostly ladders that slide back; hiding is unchanged (the pass is attacker-computable), only the honest size shrank.

Packing (the default output since this change) turns the 1.6M-cube 2-eff finals into $\sim 350k$ generalized gates. Each gate’s activation function is brought to algebraic normal form (ANF, the unique spelling of a Boolean function) and then compacted into a mixed-polarity ESOP by a deterministic procedure that reads the ANF alone, so the file still has one spelling per function, at $1.09\times$ the original cube count (the raw ANF would be $2.4\times$) and $0.95\times$ the xz-compressed bytes of the cube form. The gain is not space but the removal of representation information.

2 Background: what the old pass could not see

The old pass gathers same-target gates that can float to a common point, XOR-reduces each group (pairwise catalogue, then exact/ANF re-expression when it wins), and interleaves an adjacent

conjugation-descent (“downhill”) pass. A group closed on any read of its target and on any write to any wire it reads.

A census of the removable identical-gate pairs left in a delivered final (section “Anatomy” of `reports/collision_identities_20260905/`) showed that 99% of them are *conjugated copies*: a control u of gate g is flipped in between by a gate whose condition is implied by g ’s other literals, and the second copy reads u with the opposite polarity. From the K2 final, gates 837–839:

```
w398 ^= AND(!w146  w356 w442)
w356 ^= AND(!w146      w442)
w398 ^= AND(!w146 !w356 w442)
```

Both toggles equal $\neg w_{146} w_{356}^{\text{old}} w_{442}$; deleting the pair changes nothing. This is the three-gate Toffoli relation $g \cdot X = X \cdot g'$. The old gatherer never formed the group: the flip writes one of g ’s controls, which closed it.

3 The new compression rules

All three are on by default and each has a `-no-*` flag; with all three off the pass reproduces the 2026-08-22 output byte for byte.

3.1 Transport (Toffoli sliding at any distance)

When a gate h writes a wire u that an open group T (target t) reads, and h does not read t , the group is no longer closed: its ESOP is conjugated across h by the substitution $u \leftarrow u \oplus \text{fire}(h)$ (the downhill algebra, exact because every gate is an involution):

$$T \cdot h = h \cdot T', \quad T' = T[u \leftarrow u \oplus \text{fire}(h)].$$

For a cube $L \cdot u$ this gives $L \cdot u \oplus L \cdot \text{fire}(h)$; when $\text{fire}(h)$ ’s cube is implied by L the second term collapses to L , so $L \cdot u$ becomes $L \cdot \neg u$ and the conjugated copy of the pair above meets it and cancels. The transport is accepted when the catalogue-reduced result does not grow in (gates, literals); otherwise the group closes as before. `-transport-slack N` allows N extra cubes speculatively (measured worse; default 0).

Frames. A transported group’s cubes are written in the frame after h , so the group records a dependency on h ’s group and must be emitted after it. Closing a group closes its dependencies first; a close set is emitted in dependency order (ascending last-member order among independent groups). Two open groups with no dependency path between them commute, which makes any such order legal; the dependency graph is kept acyclic by construction (a transport that would close a cycle closes the group instead). One ordering subtlety: all candidates for a write are decided first, the refused ones are closed, and only then is the writer’s slot fixed and the accepted transports applied—a refusal-close can cascade into the writer’s own group or into an already-transported candidate, which would leave a stale frame.

3.2 Separation-aware reads

A reader of t that shares a wire with every member of the group at the opposite polarity fires on disjoint inputs from each of them, so it commutes with the whole group and the group floats past

it instead of closing.

3.3 Reversed-list gather

Each iteration also gathers the reversed gate list. Reversal is exact (every gate is an involution, so the reversed list computes the inverse function), and a forward gather of it is a leftward float of the circuit. The crossing walk floats gates both ways, so a case-split ladder left by a leftward crossing is only reachable from the left; the adjacent downhill pass sees only immediate neighbours.

3.4 Guard, determinism, verification

Gathering with transport can in principle end an iteration larger (a transported group may reduce worse than its parts would have separately), so an iteration that grows in (gates, literals) is discarded and the previous circuit kept; this fired only in the slack experiment. The output is deterministic and independent of the seed (which feeds only the verification samplers). Verification: per-group exhaustive/sampled checks, an 8×64 -lane check of every transport span, the global check of the written artifact against the input, and, for the results above, the independent evaluator.

3.5 Results and ablations

A1 benchmark, 607,365 gates, each rule removed in turn:

variant	gates out	%
all rules (default)	420,279	69.2
20 iterations instead of 10	419,411	69.1
no downhill	428,342	70.5
no separated reads	446,480	73.5
no reverse gather	491,572	80.9
transport slack 1	519,727	85.6 (regressed at iteration 4)
no transport	563,413	
old pass	575,396	94.7

Transport is the lever; the other two rules amplify it. Runtime roughly doubles (A1: 296s $\rightarrow$ 615s; the 3.7M-gate K2 final: 13 min). On the K2 final the first iteration accepted 803k transports, found 1.93M no-op ones (the group commutes with the writer), refused 773k on cost and 354 for acyclicity, and passed 574k separated readers; over ten iterations 2.9M transports were accepted and 4.7M refused. The removable-pair census on that circuit drops from 8,972 pairs (8,716 semantic-only) to 3,867 (2,144), mostly pairs the last iteration re-created.

Open leads: run on the server’s post-crossing intermediate directly; the new output holds 2,192 gates that never fire on 8,192 random inputs (old: 17), a value-level dead-gate or wire-equality lead; the refusal policy.

4 Packing

4.1 Method

At a fixed point of the gather, every maximal run of consecutive same-target gates is one gathered group: a group floats to one close point and is emitted there, and runs that could still float together would have been merged. Packing spells each run as one generalized gate $t \hat{=} f(\text{controls})$ in two steps.

1. **ANF.** f is brought to algebraic normal form, the XOR of positive monomials over the run's support, by expanding every cube (a negative literal is $1 \oplus w$, so a cube with k negative literals expands to 2^k monomials) and cancelling duplicates; a **comp** bit is the constant monomial. Any support size is handled. The ANF is the unique representation of f , so at this point nothing of the run's cube spelling survives.
2. **Compaction.** The ANF is re-expressed as a mixed-polarity ESOP by the reducer's deterministic strategies—the exact minimum tables for supports of at most 4 wires, otherwise the best of greedy subcube cover plus maximum distance-1 matching and matching alone—computed from the sorted monomial set alone, never from the cubes the ANF came from. A deterministic function of the ANF is a function of f , so the result is still one spelling per activation function; the ANF is regenerated from it whenever needed. Terms are then sorted canonically. Runs whose support exceeds the 63-wire mask width (12 per 2-eff circuit, at most 80 wires) are left in ANF, which is canonical too.

The number of packed gates equals the run count; the packed file is exact (its expansion is verified as part of **fcompress**'s global check) and, like the rest of the pass, attacker-computable.

4.2 The esop1 format

```
esop1 <wires> <gates>
<target> <n_terms> [<width> <w_1> <p_1> ... <w_width> <p_width>]*
                                     (one line per packed gate)
```

Literals are (wire, polarity 0/1) as in **mpmct1**. Wires ascend inside a term, terms are sorted by (size, literals), an empty term is the constant 1. Example: 87 3 1 423 0 2 131 1 391 1 2 254 1 302 1 is $w_{87} \hat{=} \neg w_{423} \oplus w_{131}w_{391} \oplus w_{254}w_{302}$. **format::read_mpmct** recognises the header and loads an **esop1** file as its term expansion (one cube gate per term), so every existing consumer accepts packed circuits; **format::read_packed** gives the packed gates themselves. **fcompress** writes **esop1** by default, **-no-pack** writes the **mpmct1** cube circuit, **-pack-census** prints the statistics below.

4.3 Rationale

The cube list the reducer emits is the catalogue-reduced descendant of whatever cubes the mixer left, so the same activation function comes out as different cube sets depending on the path that produced it: the spelling carries history that is not needed for evaluation and depends on the original circuit. A function-determined representation removes that channel. The ANF is the unique one that needs no algorithm to specify, but it is spelled in positive terms and costs $2.4\times$

the cubes; the compacted ESOP keeps the property (it is a fixed function of the ANF) at about the cube count, so it is the form that is kept.

What packing does not canonicalize: which gates end up in one group, where a group is emitted, the order among commuting groups and the frame a transported group is written in are all decided by the gather from the input order, which the mixer’s history determined. A fully canonical circuit would need rules for those too and is out of reach; the pipeline’s randomization is relied on for them, and packing removes the per-gate spelling only.

4.4 Measurements (compressed K2 2-eff finals)

	hold-30 2-eff	balanced 2-eff
cubes (gates before packing)	1,607,266	1,614,661
packed gates	347,055 (21.6%)	351,546 (21.8%)
ANF monomials (intermediate)	3,820,336 (2.38× cubes)	3,840,880 (2.38×)
compacted ESOP terms	1,744,830 (1.09× cubes, 0.46× ANF)	1,755,334 (1.09×)
literals, cubes → compacted	3.86M → 4.00M	3.86M → 4.01M
terms per packed gate	median 2, p90 13, p99 29, max 168	median 2, p90 13, p99 28, max 170
multi-cube runs	57.6% of runs, 90.8% of gate mass	57.1%, 90.7%
file bytes, mpmct1 → esop1	34.4 MB → 28.8 MB (0.84×)	35.1 MB → 29.2 MB (0.83×)
xz -9, mpmct1 → esop1	6.51 MB → 6.18 MB (0.95×)	6.56 MB → 6.24 MB (0.95×)
gzip -9, mpmct1 → esop1	8.25 MB → 7.55 MB	8.38 MB → 7.64 MB
(raw ANF, for reference)	41.2 MB raw, 8.58 MB xz (1.32×)	41.6 MB raw, 8.66 MB xz (1.32×)

Under xz the cube form sits close to a naive entropy count (about 32 bits per cube gate), and the compacted ESOP is slightly below it: the per-gate line overhead of 1.6M separate gates is gone and the term content is nearly the same. The compaction is smaller than the original cubes on 6.4k runs and larger on 105k runs (+144k terms), i.e. the deterministic procedure pays about 9% over the history-dependent spelling for being history-free. On the delivered K2 hold-10 final the same packing gives 449,431 packed gates for 2,165,945 cubes.

Evaluation cost. The compacted form costs about what the cube form costs: 1.09× the terms and 1.04× the literals, so about 10 ms per batch of 64 inputs on the 1.6M-cube hold-30 circuit with the compiled 64-lane evaluator (9.5 ms for the cube form). Evaluating the raw ANF instead would cost 2.3× the word operations. The canonical form fixes the description, not the evaluation strategy: an evaluator may re-derive any equivalent form once.

4.5 Deliverables

The packed forms of the two re-compressed 2-eff finals:

`circuits/bv5gss128_k2h30_2eff_final_fc2.esop1` (347,055 packed gates, 1,744,830 terms),

`circuits/bv5gss128_bal_k2_2eff_final_fc2.esop1` (351,546 packed gates, 1,755,334 terms).

Each was sanitized (header plus gate lines only, every term checked sorted, distinct and in range, terms in canonical order) and verified equal to its cube-form source by the independent evaluator on 2,048 random inputs. The pipeline (`scripts/gss_mix.sh`) writes `final.esop1` as the stage-6 deliverable and logs both the cube residual and the packed count.